

Sheng Zhang

sz696@cornell.edu

809 E. State St., Ithaca, NY, 14850

mobile: 425-666-8168

EDUCATION

Cornell University, Operation Research and Information Engineering, Ithaca, NY
Master of Engineering, Financial Engineering

December 2025

New York University, Shanghai, China
Bachelor of Science, Finance, Math, Data Science, GPA: 3.82

May 2024

Selected Coursework: Fixed Income Securities & Interest Rate Options, Econometrics, Derivatives, Volatility Modeling, Monte Carlo Simulation, Algorithmic Trading and Market Microstructure, Stochastic Calculus, Machine Learning, Optimization

SKILLS AND CERTIFICATES

- **Computer Skills:** Python, Scikit-learn, MATLAB, R, SQL, Stata, Financial Terminals (Bloomberg, Wind), Excel VBA
- **Certificates:** Bloomberg Market Concepts, WorldQuant Challenge Gold Level

PROFESSIONAL EXPERIENCE

Quant Research Intern, TruValue Asset Management, Shenzhen, China

June to Aug. 2024

- Enhanced Share Information for Stock Trend Forecasting (HIST) model to dynamically learn and adjusts pre-defined concepts and hidden concepts among stocks based on daily price volume data and industry categories.
- Constructed an adaptive neural network with Gated Recurrent Unit and Graph Attention Network (GAT) to integrate risk and fundamental factors for stock selection.
- Analyzed backtest performance of factors from RNNs, HIST, and GAT models based on excess returns, IC, and correlation.
- Integrated structured and unstructured features (risk metrics, fundamentals, industry data) into machine learning pipelines.

Quant Research Intern, Caitong Securities Asset Management, Shanghai, China

Sep. to Dec. 2023

- Programmed the Volatility Index (VIX) and SKEW Index models using Python for A-shares market volatility monitoring.
- Built a framework using Python Object-Oriented Programming to backtest high-frequency trading strategies on tick data.
- Tested statistical arbitrage trading strategies for 16 bond ETFs based on MACD, RSI, and Hurst Exponent, achieving an optimal 5% annualized return.

Business Analyst Intern, Ernst & Young (China) Advisory Ltd., Shanghai, China

June to Aug. 2023

- Managed and optimized data entries of the Tax Management System and related databases through SQL and Excel.
- Created universal Financial Statement templates for the Tax Management System using HTML and Excel VBA for automated data fetching and tax report generation.
- Validated and streamlined calculations of the Transfer Pricing model, and built pivot tables in Excel for clients.

RESEARCH EXPERIENCE

Natural Language Processing Kaggle Projects, Cornell University, Ithaca, New York

Sep. to Dec. 2024

- Designed and implemented machine learning models (Logistic Regression, Linear SVM, BERT) for multi-class text emotion classification, applying quantitative techniques to extract actionable insights from unstructured data.
- Conducted robust data preprocessing and feature engineering, including TF-IDF vectorization and lemmatization, to optimize input data for predictive modeling and improve signal-to-noise ratio.
- Performed model validation and performance analysis using quantitative metrics (accuracy, precision, recall, F1-score), systematically tuning hyperparameters to enhance out-of-sample predictive accuracy.

Financial Engineering Capstone Project, NYU, Shanghai, China

Jan. to May 2024

- Utilized high-frequency price volume data on U.S. stocks and indices to improve short-term volatility forecast accuracy during the coronavirus recession period; added extra leverage effect terms (LHAR model) to HAR Model.
- Compared the performance of HAR model with LHAR model and the RV-HAR-SV model through regression and loss function analysis; demonstrated empirical importance of incorporating the leverage effect through Diebold-Mariano Test and Ljung-Box Test.

Stochastic Processes Research Project, NYU, Shanghai, China

Sep. to Dec. 2023

- Researched the derivation of the Black-Scholes model with Geometric Brownian Motion, Ito's Lemma, and SDE.
- Implemented a Monte Carlo simulation in MATLAB to estimate option prices.

LEADERSHIP EXPERIENCE

Vice President & Co-founder, Entrepreneurs in Action, Shanghai, China

Jan. 2021 to Jan. 2022

- Coordinated Founders' Series networking events, fostering networking among NYU undergraduates and entrepreneurs.
- Developed and managed the WeChat Official Account and website, improving user engagement and online presence.
- Secured sponsorship by negotiating a recurring partnership with a local NPO.

INTERESTS

- Rubik's Cube, Power Lifting, Marathon, Table Tennis